

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I D. Arund paul (19242527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

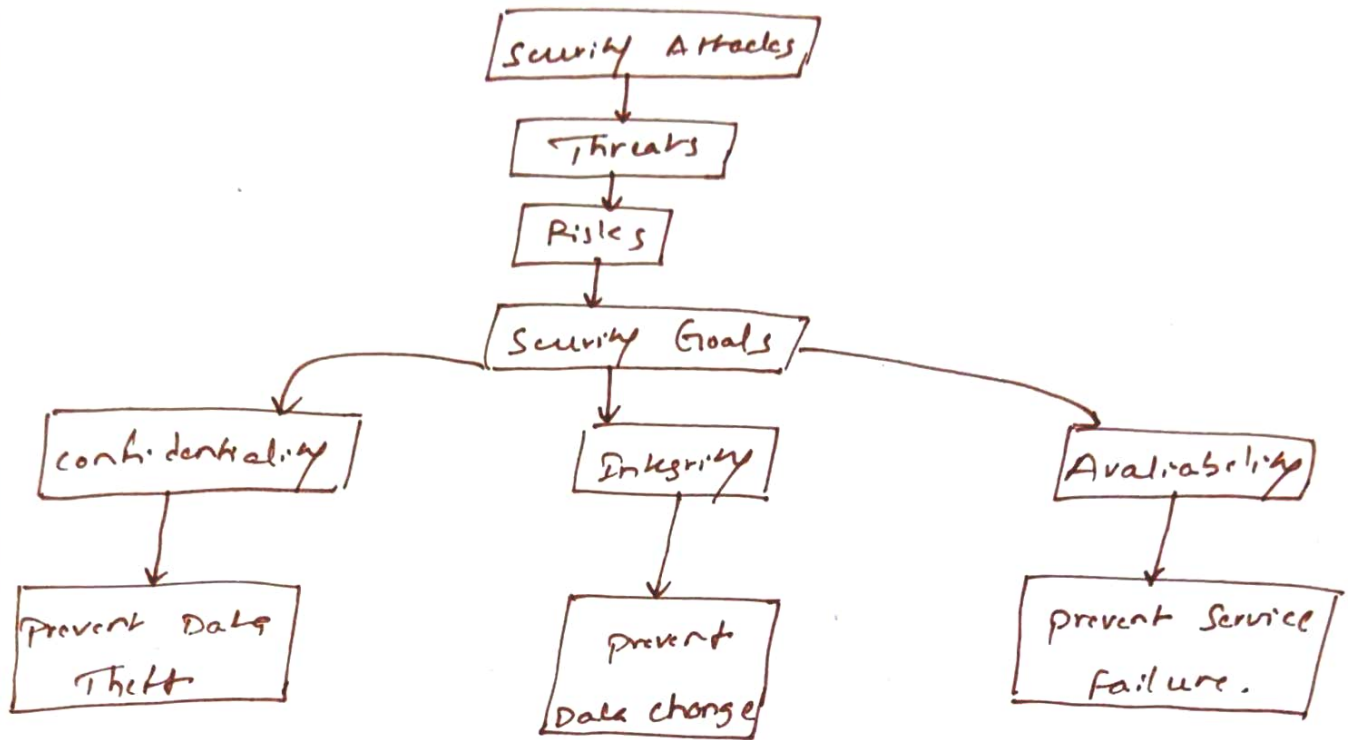


Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

Concept map



a) How attacks leads to threats and risks.

Security Attack	Threat	Risk	Goal
Phishing	stealing usernames	Data theft	confidentiality
malware	unauthorized access	Loss of info.	Confidentiality
Data Modification	Unauthorized change of info	Incorrect data	Integrity
DDos attack	making service unavailable.	Service interruption	Availability
Password attack.	Unauthorized account access.	personal data exposure.	confidentiality

b) 1. Confidentiality :-

Confidentiality ensures that information is accessible only to authorized users.

2. Integrity :-

Integrity ensures that information is accurate and has not been changed without permission.

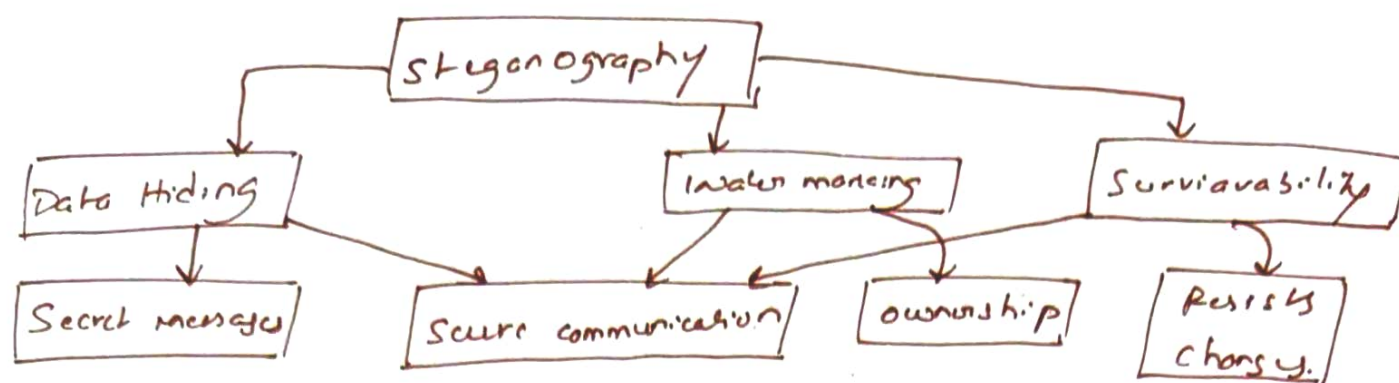
3. Availability :-

Availability ensures that systems and information are accessible whenever authorized users need them.

Conclusion :-

Security attacks can create threat that result in risks for an organization. Three major security goals - confidentiality, integrity, and availability, help reduce these risks and protect information.

4.



a) Data Hiding :-

Data Hiding is the main concept of steganography. It means placing secret information inside another normal-looking medium.

Watermarking :-

Watermarking is a special form of information embedding. It is mainly used to identify ownerships, prove authenticity or protect digital content.

Survivability :-

Survivability refers to the ability of hidden information or a watermark to remain present and usable even after operations such as compression, resizing.

Data Hiding $\rightarrow$ Watermarking $\rightarrow$ Survivability.

b) Data Hiding $\rightarrow$ confidentiality

* watermarking $\rightarrow$ Authenticity and ownership.

* Survivability $\rightarrow$ Reliability

Concept.
Conclusion

Purpose

Real world.

c) Data Hiding $\rightarrow$ Hide Secret info $\rightarrow$ confidential message hidden inside image

watermarking $\rightarrow$ protect ownership $\rightarrow$ copyright watermark on photos.

Survivability $\rightarrow$ present hidden data after modifications $\rightarrow$ watermark remaining after JPEG.

Conclusion:-

Steganography provides a way to hide information with ordinary digital media. Data hiding provides secrecy, watermarking provides ownership and authenticity and Survivability ensures that the embedded information remains effective after common modifications.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation